ALEJANDRO ORTIZ PERDOMO

MLOps Engineer | Machine Learning Engineer

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alejortizp.github.io/curriculum_HV

PROFESSIONAL SUMMARY

MLOps Engineer focused on taking ML models and LLMs **from data to production**: data pipelines, reproducible training, deployment and monitoring. Implemented **CI/CD** with **Databricks + MLflow** that cut deployment from 2 weeks to **2 days**, and **data/model drift** monitoring with automated retraining. Experience deploying services on **Azure, AWS and GCP** with **Docker**, **Kubernetes (EKS)**, **Terraform** and serverless (**Azure Functions**, **Lambda**, **Cloud Run**), automated with **GitHub Actions**.

WORK EXPERIENCE

AI Engineer — AI Practice
Jun 2025 – Present

Periferia IT Group

- **Production CV-Job semantic search pipeline:** Took to production the ingestion (**Azure Document Intelligence**) and vector indexing (**Azure AI Search**) pipeline, packaged as **Dockerized Azure Functions** with SQL-triggered incremental indexing and scheduled sync. Processes **500+ CVs/day** and serves **10k+ searches/month** at **<200ms** latency.
- **Multi-cloud LLM agent platform (SRPA):** Technical design and documentation of a *plug-and-play* agent platform automating the SDLC, with LLMs such as **DeepSeek R-1** served on **Azure AI Foundry** and run on **Azure / AWS / GCP** for internal and commercial use; includes automated unit-test generation with ≥**85%** coverage.
- **Production reference architectures:** Define architectures, estimates and team sizing to take **GenAI, Agentic AI and RAG** solutions to production for enterprise clients (banking, insurance, retail, energy and telco).

EDUCATION

Master in Software Engineering
Universidad de los Andes
2025 – In Progress

B.S. in Statistics
Universidad del Valle
2011 – 2016

Data Science Diploma (DS4A)
Correlation One, 2022

Deep Learning Diploma
Univ. Autónoma Occidente, 2023

LANGUAGES

Spanish
Native

English
Professional Proficiency (B2)

KEY SKILLS

Effective Communication
Problem Solving
Analytical Thinking
Team Collaboration
Adaptability
Continuous Learning
Time Management
Technical Leadership

Freelance AI Engineer — Dental Clinic
Jul – Aug 2026

Freelance

- **Serverless LLM agent deployment:** Took two agents (**LangGraph** + **Azure OpenAI**) to production on **Azure Functions** with queue-based processing (poison-queue handling and deduplication), Table Storage persistence and **Key Vault** secrets via managed identity.
- **Integrations & observability: WhatsApp Cloud API** webhook with HMAC verification, **Outlook (Microsoft Graph)** calendar with retries and token rotation, and **Application Insights** alerting. Deterministic critical logic (locking against double booking) and test-driven development with **118 tests**.

Machine Learning Engineer
Sep 2023 – Jun 2025

Global Hitss

- **End-to-end MLOps on Databricks:** Took models from data to production: **PySpark** processing on **Delta Lake** governed by **Unity Catalog**, orchestration with **Databricks Workflows**, versioning and promotion in the **MLflow Model Registry**, and **endpoint** or **batch** serving depending on the use case.
- **Model CI/CD & operations:** Implemented the CI/CD pipeline that cut deployment from 2 weeks to **2 days**, with automated drift monitoring and retraining.
- **Production RAG agent:** Deployed a RAG conversational agent (**LangChain**, **Azure OpenAI**, **FAISS**) serving **10k+ queries/month**.
- **Models in production:** Productized the *Next Best Offer* system (Reinforcement Learning) with rollout validated by **A/B testing**, and the churn model with scores fed into the **CRM** for **50k+ customers**.

Machine Learning Consultant
Aug – Sep 2023

Grupo CINTE

- **ML Observability:** Designed the monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Inference optimization:** Audited and optimized LightGBM and Random Forest models, reducing inference times and compute cost.

Data Scientist
Mar 2023 – Jul 2023

NAE

- **Contactability model in operation:** Built and put into operation a **LightGBM** model prioritizing time windows and channels for campaigns of **100k+ contacts/month**.

CERTIFICATIONS

AI-102: Microsoft Certified Azure AI Engineer Associate
Microsoft, Feb 2026

NLP Specialization (RAG/Transformers)
Coursera, Jun 2025

PyTorch Ultimate
Coursera, Mar 2025

MLOps and Databricks
Netec, Dec 2024

AWS Cloud Practitioner
Udemy, Feb 2022

TECHNICAL SKILLS

(years exp.)

GENAI & LLMS

LangChain 2+ (Agents, RAG) | **LangGraph** 1+

Azure OpenAI 2+ (GPT-4, Embeddings)

Prompt Engineering 2+

Vector DBs 2+ (FAISS, Qdrant, Pinecone)

LLM Evaluation 1+ (RAGAS)

AZURE AI & CLOUD

Azure AI Search 1+

Document Intelligence 1+

Azure AI Foundry 1+ | **Azure Functions** 1+

Databricks + MLflow 3+ (Workflows, Model Registry)

Delta Lake 1+ | **Unity Catalog** 1+

GOOGLE CLOUD PLATFORM (GCP)

Vertex AI 1+ (Training, Endpoints, Pipelines)

Cloud Run 1+ | **Pub/Sub** 1+

Compute Engine 1+ (MIG zonales)

Cloud Storage 1+ | **Cloud SQL** 1+

AMAZON WEB SERVICES (AWS)

EKS 1+ | **Lambda** 1+ | **DynamoDB** 1+

SNS 1+ | **SQS** 1+ | **RDS** 1+

ML/DL FRAMEWORKS

PyTorch 2+ | **TensorFlow** 3+

Scikit-learn 4+ | **XGBoost** 4+ | **LightGBM** 4+

Hugging Face Transformers 2+

Data Scientist II

May 2022 – Jan 2023

Xolit

- **Batch production deployment:** Took models to production as **batch** jobs orchestrated with **Apache Airflow** on a defined schedule, delivering scores to the collections operation.
- **Production model monitoring:** Built **Power BI** dashboards tracking production model performance metrics and collection KPIs.
- **Models in operation:** Payment propensity (feeding **automated management priorities** for the portfolio), optimal contact time-slot recommendation and clustering-based account-holder segmentation.

OPEN SOURCE PROJECTS

🏦 RAG Assistant with Web Scraping and RAGAS Evaluation

— github.com/alejortizp/rag-banco

Conversational assistant over a bank's public website: robots.txt-compliant crawler, **Qdrant** indexing, cross-encoder reranking and per-session memory. Evaluated with **RAGAS** (50 questions, independent LLM judge): faithfulness **0.87**; the reranker improved context utilization by **+8 pts** (A/B).

Stack: Qdrant, Sentence-Transformers, Cross-Encoder, Groq Llama 3.1, RAGAS, Streamlit, Docker

🤖 Email Agent with LangChain

— github.com/alejortizp/email-agent

Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.

Stack: LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering

💬 NL-to-SQL/Pandas Agent

— github.com/alejortizp/nl-to-code

Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.

Stack: LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern

ACADEMIC PROJECTS — M.SC. (UNIANDES)

☁️ Teaching-Assistant Platform on GCP (Go + Angular)

MISW4204 Cloud Solutions Development · 2026

Top contributor (**60%** of commits, team of 4). Migrated the system from Docker Compose to an **autoscaling MIG + Load Balancer** and then to serverless **Cloud Run + Pub/Sub**, with asynchronous AI-generated PDF reports. JMeter load tests: **158 rps** (p95 292 ms) with **0%** infrastructure errors; **84%** (Go) and **93%** (Angular) coverage; SonarQube A rating.

Stack: Go (Gin), Angular, PostgreSQL, Cloud Run, Pub/Sub, Cloud SQL, OpenAI, JMeter, SonarQube, GitHub Actions

MLOPS & ENGINEERING

Python 7+ | **SQL** 7+ | **PySpark** 3+
Docker 2+ | **CI/CD** 2+
Apache Airflow
Kubernetes 1+ | **Terraform** 1+
Go 1+ | **Angular** 1+ | **Kotlin** 1+
FastAPI 3+ | **Streamlit** 3+
Git 4+ | **GitHub Actions** 2+

🚚 Event-Driven Delivery Marketplace on AWS

MISW4301 Cloud-Native App Development · 2026

Led the AWS architecture and IaC for **11 FastAPI microservices** (hexagonal architecture) on **EKS**, with a serverless extension: **SNS→SQS** with DLQs, **4 container Lambdas** and DynamoDB, all in **Terraform** deployed via GitHub Actions to 5 environments. Fixed cold starts and IAM permissions, reaching **76/76** E2E assertions with zero failures.

Stack: AWS (EKS, Lambda, SNS, SQS, DynamoDB, RDS), Terraform, Kubernetes, FastAPI, Docker

🛡️ Microservices Architecture Experiments

MISW4202 Agile Software Architectures · 2026

Hypothesis-driven experiments for a booking system: asynchronous payment-anomaly detection in **538 ms** mean vs. a 2 s target (t-test, $p<0.001$) and **100%** Man-in-the-Middle attack detection with zero false positives (RS256 JWT with revocation, RSA-OAEP, integrity hashing, TLS).

Stack: Flask, Redis Streams, Celery, Nginx, JWT, RSA, SciPy

⏰ QR Alarms: from UX Research to Android + Web App

— github.com/a-ortizp/alarmas-qr-app

UX Prototyping · MISO · 2026

Full product cycle: user research, **3 usability tests** (20 findings), Figma prototype and Material 3 design system (48 components); then a **22-screen** Android app (ML Kit QR scanning, alarms firing with the app closed) and Angular web app, with signed APK releases via GitHub Actions.

Stack: Figma, Kotlin, Jetpack Compose, CameraX, ML Kit, Angular, GitHub Actions